

1 **Data release - "2022_Q4_IBL_et_al_BWM" - Technical White Paper**

2 International Brain Laboratory

3 **Contents**

4	Introduction	2
5	Recording strategy	2
6	Data set overview	4
7	Data access	6
8	Contact details	6
9	Methods	7
10	Overview	7
11	Animals	8
12	Headbar implant surgery	8
13	Materials and apparatus	8
14	Habituation, training, and experimental protocol	8
15	Electrophysiological recording using Neuropixels probes	9
16	Serial section two-photon imaging	9
17	Probe track tracing and alignment	9
18	Spike sorting	10
19	Coverage	10
20	Data quality	11
21	Sessions and insertions	11
22	Trials	11
23	Neurons and brain regions	11
24	Wheel data	11
25	Video data	11
26	Raw video QC	13
27	Pose estimation QC	13
28	Electrophysiology data	16
29	Raw ephys data QC	16
30	Spike sorting QC	16
31	Acknowledgments	28
32	Diversity Statement	28

Introduction

The International Brain Laboratory (IBL; www.internationalbrainlab.org) is a collaboration of 22 labs with the shared mission of advancing our understanding of the neural systems and circuits that underlie behavior. The IBL approaches this challenge with a distributed experiment model, in which scientists within each lab contribute to the same shared scientific experiment. Data collection takes place across 12 labs and the data are preprocessed and stored centrally. The collaboration emphasizes an open science ethic in which data and code are shared freely.

Here, we present our first release of data from the "Brain-wide Map" project, in which we have systematically recorded from nearly all major brain areas with Neuropixels probes. Our recording strategy, described in detail below, used a grid system for unbiased sampling and replicated each recording site in at least two laboratories. These data constitute a brain-wide map of activity at single-spike cellular resolution during a decision-making task. In addition to the map, this data set contains other information gathered during the task: sensory stimuli presented to the mouse; mouse decisions and response times; and mouse behavioral information from analyzed video recordings.

This is the first release of a living dataset: additional recordings are ongoing and future data releases will increase the size of the dataset.

Recording strategy

Our recording strategy aimed to achieve these goals:

1. Map the activity from all brain regions, spanning from rostral tip of the left forebrain to the right brainstem and cerebellum caudally. However, we excluded the olfactory bulb region.
2. Obtain a uniform coverage across the brain volume
3. Assess the reproducibility of recordings across subjects and laboratories (via the "repeated site")
4. Assess bilateral, interhemispheric interactions (via the "bilateral sites")

To accomplish this, we designed a grid of recording sites with 500 μm spacing (Fig 1a-c). The grid locations were sampled with one of two probes per session (typically, one coming from the left of the subject and one from the right). The grid is focused on the left hemisphere to maximize the proportion of pairs of areas that share anatomical connections (for this reason, cerebellum and medulla sampled from the right hemisphere instead of left, since connections from these areas to the rest of the brain cross the midline). To assess the reproducibility of our physiological measurements across subjects and laboratories, we selected one of these recording sites as the "repeated site" (Fig 1d) and targeted a recording to this site once in every mouse of the dataset (*The International Brain Laboratory, 2022a*). Recordings from the "repeated site" sampled activity in the visual cortex, hippocampus and thalamus. To test for hemisphere-specific activity, three locations were chosen to be recorded bilaterally (i.e. left and right hemisphere of the same coordinate simultaneously) in a small number of subjects (Fig 1e). The first location sampled the visual, retrosplenial cortices areas, midbrain and pons ; the second location was the "repeated site" ; the third location sampled the motor, orbital, and olfactory cortical areas.

This grid served as an initial target; more insertions are being performed, in particular to reach areas that are currently lacking coverage. Such additional datasets will be released in coming years once processing is completed.

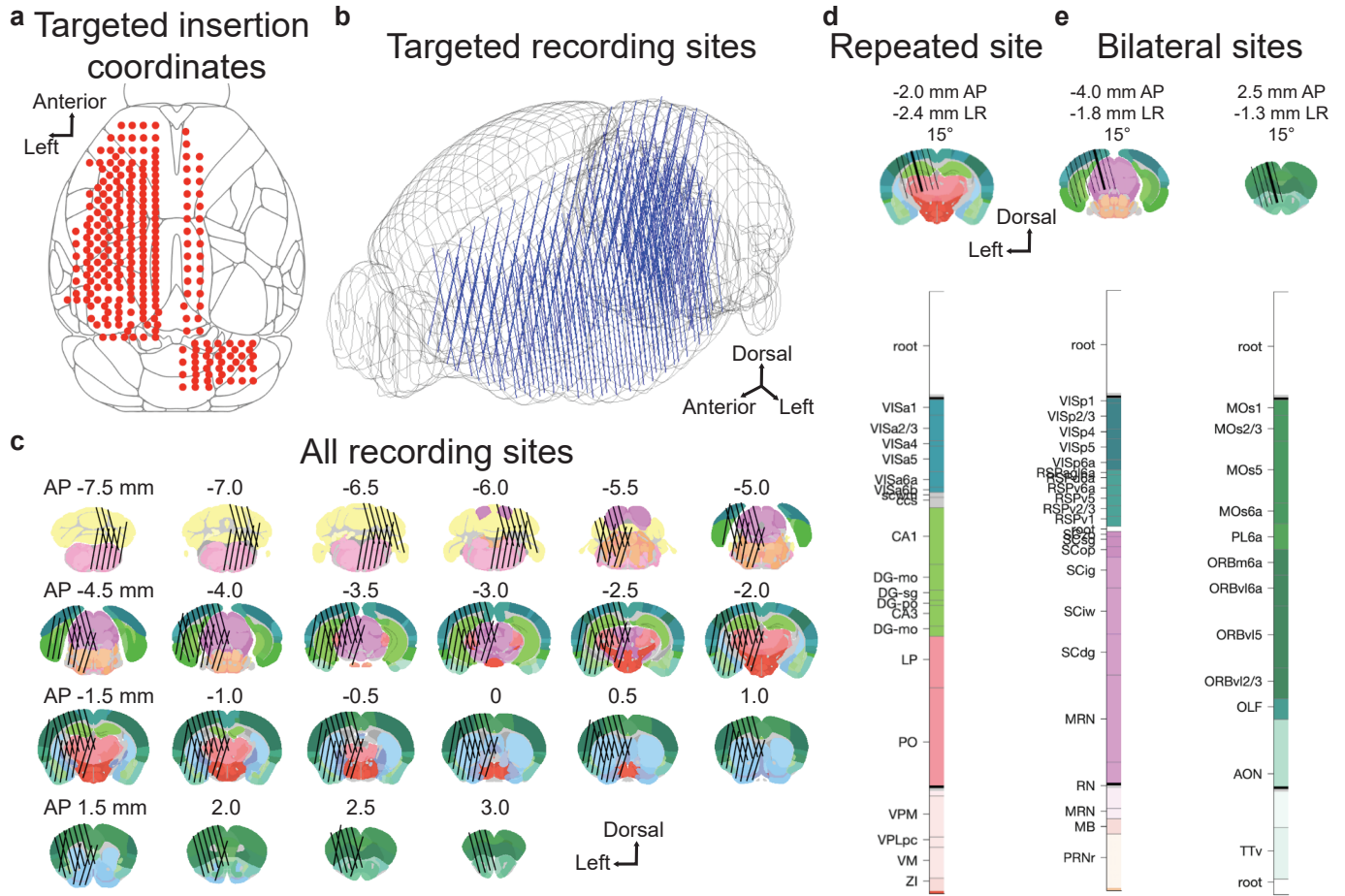


Figure 1. Recording strategy. **a.** Targeted insertion locations on the brain surface. **b.** 3D positions of all recording sites. Each line represents the 3.84 mm span of electrode locations recorded by a single probe. **c.** Exact locations of all targeted recording sites (black lines) shown on coronal slices of the Allen CCF atlas. **d.** The "Repeated Site" location. Top, A coronal slice of the brain at -2.0 mm ap with the depicted penetration highlighted in dark black. Bottom, The brain regions traversed along this trajectory, as well as those 1 mm above and below, for context. The recorded area is anticipated to be the bright area between the two solid black lines, i.e. from the top of VISa to near the bottom of PO, but not including VPM, etc. **e.** The "Bilateral Site" locations. Conventions as in d. In addition to these two sites, the Repeated Site was also recorded bilaterally.

filter	remaining			
	sessions	probes	neurons	regions
session and insertion QC	459	699	621,733	281
reaction time & missing events	459	699	621,733	281
minimum 3 error trials	459	699	621,733	281
single unit QC	459	698	75,708	268
gray matter regions	459	698	65,336	266
minimum 5 units/region	455	691	63,357	245
minimum 2 sessions/region	454	690	62,990	210
minimum 20 units/region	454	689	62,857	201

Table 1. Session and probe filtering. The table indicates the progressive filtering of the sessions and probes based on the various inclusion criteria described in the text.

Data set overview

The brain-wide-map dataset of IBL consists of 621,733 units (see section *Data quality* for definition), recorded using Neuropixels probes in 139 mice doing the IBL task, distributed across 12 labs in Europe and the US. The neural data is obtained from at most two Neuropixels probes recorded simultaneously per behavioral session (N session = 459), with each mouse undergoing multiple of such sessions, totalling to 699 insertions covering most of the left hemisphere (Fig2a).

Though recordings were targeted to an evenly spaced grid (Fig 1), differences between the subject's brain and the atlas space, along with experimental variability, led to actual recording locations as reconstructed with histology (see section *Methods*) being at slightly different locations. Accounting for the true location of each recording, our coverage in this dataset is 94% of the total targeted brain volume.

After spike sorting and single unit quality control (*The International Brain Laboratory, 2022b*), the brain region of each recorded neuron was identified. We consider a brain region to have been recorded in this dataset when at least two insertions, which each contained at least 5 neurons, were completed. By this definition, we recorded from 210 distinct parcellated regions in the Allen CCF (*Wang et al., 2020*) (see [online table](#) for details). Fig 2b shows the distribution of recorded neurons (i.e. good units) across brain regions, with horizontal black lines in the bars separating different recordings. Table 1 summarizes the filtering of sessions, recordings and putative units.

The decision making task (Fig2c,d) requires the mouse to move the visual stimulus 35° to the center of the screen, by moving a wheel with its two front paws (see section *Methods* for details). The stimulus is presented on the left or right side of the screen with a probability that varies from one block of trials to the next. Of interest, zero-contrast stimuli are presented within these blocks, which when responded to, offer an insight on the mouse prior to the likely side of the stimulus.

In order to study neural correlates of the behavior, we recorded both behavioral and neural data simultaneously. The major behavioral events of interest for analysis are the stimulus onset time, the animal's choice, the feedback delivery time, the feedback type (reward vs error tone), the probability of the stimulus being on the left (block type), and the wheel motion onset time. Of note, in this task the time between stimulus onset and wheel motion onset - the reaction time of the animals - is short, with a median of 150 ms (inset in Fig2d).

Besides these scalars describing the events ongoing in the trials, there are continuous recordings of the mice' behavior using the wheel, as well as features extracted from videos of the animals (*The International Brain Laboratory, 2021b*). Fig2e illustrates three such continuous behavioral time series, namely: the wheel speed, the whisker pad motion energy, and licking (obtained from tongue tracking on the video images), aligned to neural activity recorded in various brain regions simultaneously.

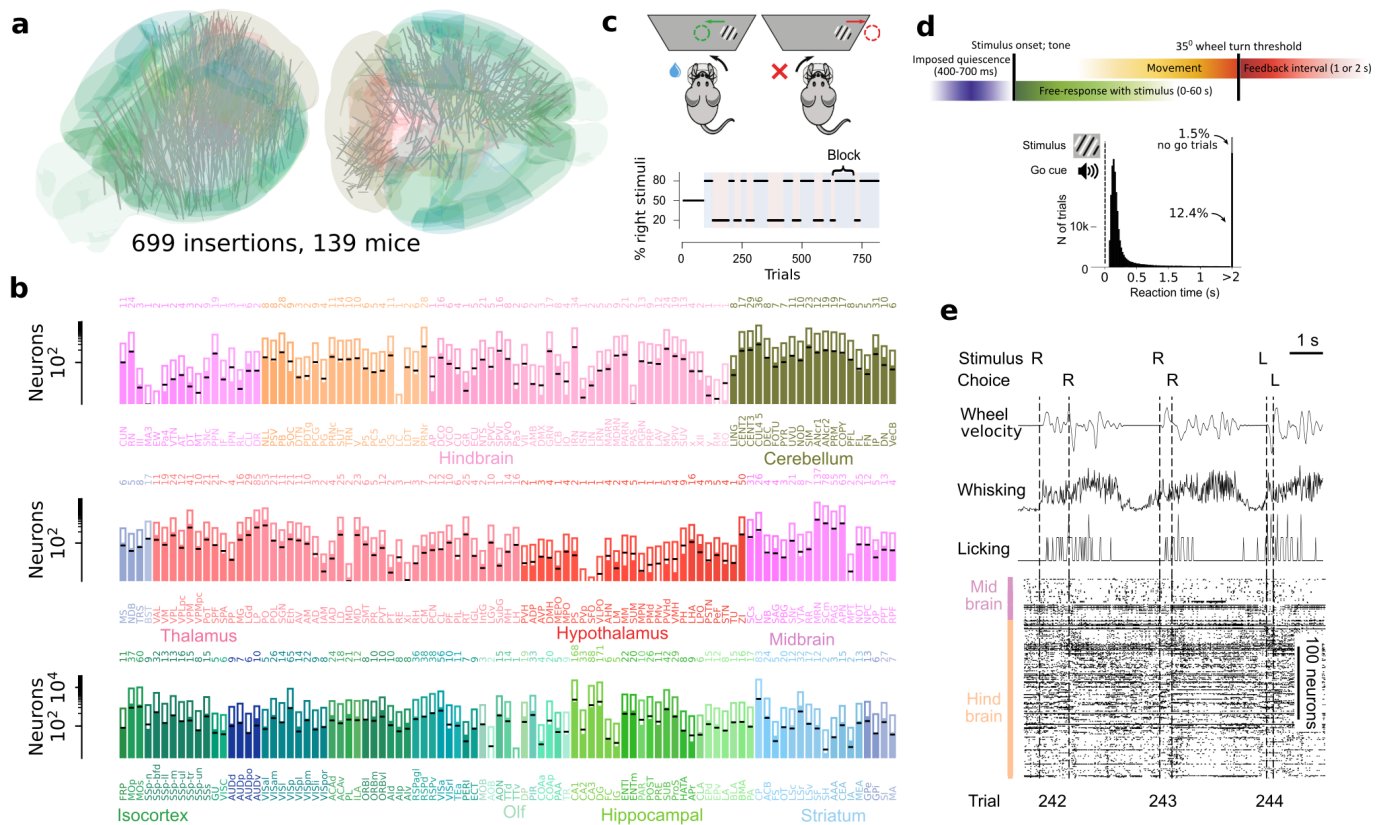


Figure 2. Task and data types. **a.** Neuropixel probe trajectories shown within a 3D brain schematic. A mesh of recordings covers most of the left hemisphere. **b.** For each region, the number of neurons recorded (full bar length) and the number of good neurons used for analysis (filled portion; for reference, the black line on each bar shows 10% of the number of recorded neurons, which is the average number of neurons that were good). Definitions of the acronyms for brain regions are available in an [online table](#). The same table reports the so-called Cosmos hierarchical grouping of the regions, e.g. 'Isocortex'. **c.** The IBL task schematic and block structure of an example session. The probability of the stimulus being on the left side is varied in blocks of consecutive trials. **d.** Temporal details of the IBL task with histogram of reaction times. **e.** Example time series and key trial information for 3 trials, resulting from the rotary encoder of the wheel, video analysis and electrophysiological recordings.

Data access

- The data collected can be viewed at <https://viz.internationalbrainlab.org/app>.
- To download the data, see the instructions at https://int-brain-lab.github.io/iblenv/notebooks_external/data_release_brainwidemap.html.
- To assess the quality of the data, see the [general QC spreadsheet](#) (see section **Data quality** for definitions)

Contact details

- Issues with the data? Post a question on one of these platforms:
 - <https://github.com/int-brain-lab/iblenv/issues>
 - <https://neurostars.org/>; make sure to apply the tag `ibl`
- General questions about the datasets or publications? Email info@internationalbrainlab.org

Methods

All procedures and experiments were carried out in accordance with the local laws and following approval by the relevant institutions: the Animal Welfare Ethical Review Body of University College London; the Institutional Animal Care and Use Committees of Cold Spring Harbor Laboratory, Princeton University, University of Washington, and University of California at Berkeley; the University Animal Welfare Committee of New York University; and the Portuguese Veterinary General Board.

Overview

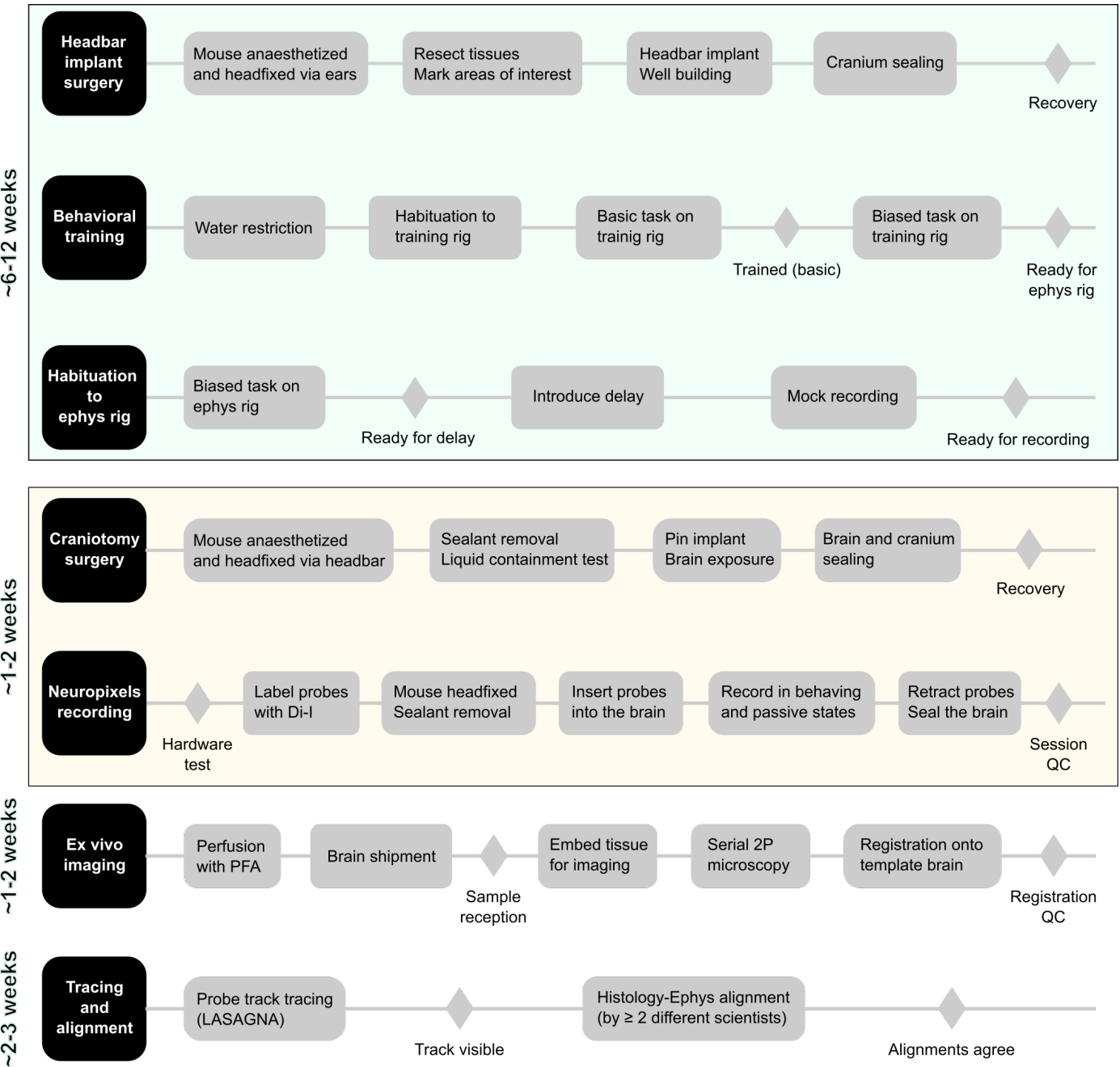


Figure 3. Overview of the experimental pipeline. The experiment follows the main steps indicated in the left-hand black squares in chronological order from top to bottom. Within each main step, actions are undertaken from left to right; diamond markers indicate points of control. Time estimate for each step of the pipeline is indicated on the left.

111 **Animals**

112 Mice were housed under a 12/12 h light/dark cycle (normal or inverted depending on the laboratory) with food and water
113 available ad libitum, except during behavioural training days. Electrophysiological recordings and behavioural training were
114 performed during either the dark or light phase of the cycle depending on the laboratory. N=139 adult mice (C57BL/6, male and
115 female, obtained from either Jackson Laboratory or Charles River) were used in this study. Mice were aged 13-173 weeks (mean
116 44.35 weeks, median 27.0 weeks) and weighed 16.1-35.7 g (mean 23.84 g, median 23.9 g) on the day of electrophysiological
117 recording.

118 **Headbar implant surgery**

119 A detailed account of the surgical methods for the headbar implant is in Appendix 1 of (*The International Brain Laboratory, 2021c*).

120 Briefly, mice were anesthetized with isoflurane and head-fixed in a stereotaxic frame. The hair was then removed from
121 their scalp, much of the scalp and underlying periosteum was removed and bregma and lambda were marked. The head was
122 positioned in stereotaxic coordinates. The head bar was then placed in one of three stereotactically defined locations and
123 cemented in place. The location of the future craniotomies were measured using a pipette referenced to Bregma, and marked on
124 the skull using either a surgical blade or pen. The exposed skull was then covered with cement (Vetbond) and clear UV curing
125 glue (Norland Optical Adhesives), ensuring that the remaining scalp was unable to retract from the implant.

126 **Materials and apparatus**

127 For detailed parts lists and installation instructions, see Appendix 1 of (*The International Brain Laboratory, 2022a*).

128 Briefly, each lab installed a standardized electrophysiological rig, which differed slightly from the apparatus used during
129 behavioral training (*The International Brain Laboratory, 2021c*). The general structure of the rig was constructed from Thorlabs
130 parts and was placed on an air table (Newport, M-VIS3036-SG2-325A) surrounded by a custom acoustical cabinet, which design
131 varied from lab to lab. A static head bar fixation clamp and a 3D-printed mouse holder were used to hold a mouse such that its
132 forepaws rested on the steering wheel (86652 and 32019, LEGO) (*The International Brain Laboratory, 2021c*). Silicone tubing
133 controlled by a pinch valve (225P011-21, NResearch) was used to deliver water rewards to the mouse. The display of the visual
134 stimuli occurred on a LCD screen (LP097Q × 1, LG). To measure the precise times of changes in the visual stimulus, a patch of
135 pixels on the LCD screen flipped between white and black at every stimulus change, and this flip was captured with a photodiode
136 (Bpod Frame2TTL, Sanworks). Ambient temperature, humidity, and barometric air pressure were measured with the Bpod
137 Ambient module (Sanworks), wheel position was monitored with a rotary encoder (05.2400.1122.1024, Kubler). Videos of the
138 mouse were recorded from 3 angles (left, right and body) with USB cameras (CM3-U3-13Y3M-CS, Point Grey) sampling at 60, 150,
139 30 Hz respectively (for details see Appendix 1 of (*The International Brain Laboratory, 2022a*)). A custom speaker (Hardware Team
140 of the Champalimaud Foundation for the Unknown, V1.1) was used to play task-related sounds, and an ultrasonic microphone
141 (Ultramic UM200K, Dodotronic) was used to record ambient noise from the rig. All task-related data was coordinated by a Bpod
142 State Machine (Sanworks). The task logic was programmed in Python and the visual stimulus presentation and video capture
143 was handled by Bonsai (*Lopes et al., 2015*) and the Bonsai package BonVision (*Lopes et al., 2021*).

144 All recordings were made using Neuropixels 1.0 probes (Imec (*Jun et al., 2017*); some probes were 3A and 3B prototype
145 models), advanced in the brain using a micromanipulator (Sensapex, uMp-4) tilted at a 15 degree angle from the vertical line.
146 Data were acquired via an FPGA (for 3A probes) or PXI (for 3B and 1.0 probes, National Instrument) system using SpikeGLX, and
147 stored on a PC.

148 **Habituation, training, and experimental protocol**

149 For a detailed protocol on animal training, see section *Methods* in (*The International Brain Laboratory, 2021c, 2022a*).

150 All the data analysed in this study were acquired when the mouse performed the task named "ephysChoiceWorld". Briefly, at
151 the beginning of each trial, the mouse was required to not move the wheel for a quiescence period of 400–700 ms. If the wheel
152 moved during this period, the timer was reset. After the quiescence period, a visual stimulus appeared on either the left or right
153 ($\pm 35^\circ$ azimuth) of the screen, with a contrast randomly selected from a predefined set. Simultaneously, an onset tone (5 kHz sine

wave) was played for 100 ms. When the stimulus appeared, the mouse had 60 s to move the wheel. A response was registered if the center of the stimulus crossed the $\pm 35^\circ$ azimuth line from its original position. If the mouse correctly moved the stimulus 35° to the center of the screen, it immediately received a 1.5 μ L reward; if it incorrectly moved the stimulus 35° away from the center, it received a timeout. If the mouse responded incorrectly or failed to reach either threshold within the 60 s window, a noise burst was played for 500 ms and the inter-trial interval was set to 2 s. In trials where the visual stimulus contrast was set to 0, the mouse had to respond as for any other trial, i.e. by turning the wheel in the correct direction in order to receive a reward - however, in such a case the mouse was not able to perceive whether the stimulus was presented on the left or right side of the screen.

Electrophysiological recording using Neuropixels probes

For details on the craniotomy surgery, see Appendix 3 of (*The International Brain Laboratory, 2022a*).

Briefly, upon the first day of electrophysiological recording, the animal was anaesthetised using isoflurane and surgically prepared. The cement and glue were removed, exposing the skull over both hemispheres. A test was made to check whether the implant could hold liquid, and if successful a grounding pin was implanted. One or multiple craniotomies (approximately 1×1 mm) were made over the marked locations. The dura was left intact, and the brain was lubricated with ACSF. DuraGel was applied over the dura as a moisturising sealant, and covered with a layer of Kwikcast. The mouse was administered with analgesics subcutaneously, and left to recover in a heating chamber until locomotor and grooming activity were fully recovered.

After a further recovery period of at least 2 hours, subjects could be head-fixed in the apparatus for recording. Once a craniotomy was made, up to 4 subsequent recording sessions were made in that same craniotomy. Once the first set of craniotomy was fully recorded from, a subject could undergo another craniotomy surgery in accordance with the institutional licence. Up to two probes were inserted in the brain on a given session.

Serial section two-photon imaging

Mice were given a terminal dose of pentobarbital and perfuse-fixed with PBS followed by 4% formaldehyde solution (ThermoFisher 28908) in 0.1M PB pH 7.4. Whole mouse brain was dissected, and post-fixed in the same fixative for a minimum of 24 hours at room temperature. Tissues were washed and stored for up to 2-3 weeks in PBS at 4C, prior to shipment to the Sainsbury Wellcome Centre for image acquisition. For full details, see Appendix 5 of (*The International Brain Laboratory, 2022a*).

For imaging, brains were equilibrated with 50mM PB solution and embedded into 5% agarose gel blocks. The brains were imaged using serial section two-photon microscopy (*Ragan et al., 2012; Economo et al., 2016*). The microscope was controlled with ScanImage Basic (Vidrio Technologies, USA), and BakingTray, a custom software wrapper for setting up the imaging parameters (*Campbell, 2020*). Image tiles were assembled into 2D planes using StitchIt (*Campbell, 2021*). Whole brain coronal image stacks were acquired at a resolution of $4.4 \times 4.4 \times 25.0$ μ m in XYZ, with a two-photon laser wavelength of 920 nm, and approximately 150 mW at the sample. The microscope cut 50 μ m sections but imaged two optical planes within each slice at depths of about 30 μ m and 55 μ m from the tissue surface. Two channels of image data were acquired simultaneously using multialkali PMTs ('Green' at 525 nm $\pm$ 25 nm; 'Red' at 570 nm low pass).

Whole brain images were downsampled to 25 m XYZ pixels and registered to the adult mouse Allen common coordinate framework (*Wang et al., 2020*) using BrainRegister (*West, 2021*), an elastix-based (*Klein et al., 2010*) registration pipeline with optimised parameters for mouse brain registration. For full details, see Appendix 7 of (*The International Brain Laboratory, 2022a*).

Probe track tracing and alignment

Neuropixels probe tracks were manually traced to yield a probe trajectory using Lasagna (*Campbell et al., 2020*), a Python-based image image viewer equipped with a plugin tailored for this task (see Appendix 6 of (*The International Brain Laboratory, 2022a*)). Traced probe track data was uploaded to an Alyx server (*Rossant et al., 2021*); a database designed for experimental neuroscience laboratories. Neuropixels channels were then manually aligned to anatomical features along the trajectory using electrophysiological landmarks with a custom electrophysiology alignment tool developed by (*Faulkner, 2020*). Another example use case of the alignment tool can be found in (*Liu et al., 2021*).

Spike sorting

The spike sorting pipeline used at IBL is described in detail in (*The International Brain Laboratory, 2022b*). Briefly, spike sorting was performed using a modified version of the Kilosort 2.5 algorithm (*Steinmetz et al., 2021*). We found it necessary to improve the original code in several aspects (scalability, reproducibility, and stability, as discussed in (*The International Brain Laboratory, 2022a*)), and developed an open-source Python port; the code repository is here: (*The International Brain Laboratory, 2021a*).

Coverage

The coverage has been evaluated by taking a cylinder of radius 354 μm around each probe insertion. Voxels that fall within this cylinder are denoted as “recorded”. We iterated through all insertions in our Brainwide map recordings to evaluate how many voxels have been targeted by one or more insertions. We recorded 94% of the targeted volume with at least 1 probe and 87 % with at least 2 probes.

Data quality

Sessions and insertions

Electrophysiology sessions were included if the mice performed at least 400 trials, with a performance of at least 90% correct on easy contrasts (across block types).

Also, sessions were included if they reached a set threshold on defined hardware tests (definitions can be found [here](#)). Sessions for which less than 200 trials passed trial exclusion criteria (see below) are released but marked as 'FAIL' in the *TrialEvents* column of this [general QC spreadsheet](#).

Insertions were disregarded if the raw neural data presented major artifacts, or if the probe tract could not be recovered during the histology procedure. Furthermore, insertions are included only if their alignments have been resolved (see Appendix 6 of (*The International Brain Laboratory, 2022a*) for definition).

Trials

The following trial exclusion criteria were considered: Any of the following trial events could not be detected, and hence is set to NaN: `choice`, `probabilityLeft`, `feedbackType`, `feedback times`, `stimOn times`, `firstMovement times`. The difference between `firstMovement times` and `stimOn times` is not in the range of 0.08 and 2.0 seconds. Note that the data for trials not passing these criteria is still released. The exclusion criteria were solely used to assess which sessions are marked as 'FAIL' (see above).

Detailed trials quality control outputs and the associated code can be found in the associated [github repository](#).

Neurons and brain regions

Units output by the spike sorting pipeline were marked as bad units if they failed one of the three criteria described in (*The International Brain Laboratory, 2022b*): median amplitude > 50 uV ; noise cut-off < 5 ; refractory period violation. Units that pass all the three criteria are considered good for analysis and are termed "neurons" in this document. Note that we computed the single unit quality metrics for every single recording session, regardless of the raw ephys or spike sorting quality outcomes (described in sections below). Out of the 621,733 units collected, 75,708 passed the quality control. We released all units as part of the data release. Instructions on how to filter for good units upon data downloading can be found [at this url](#).

Brain regions were included in the final count only if they had at least 2 included sessions and at least 10 neurons.

Wheel data

For the wheel position data derived from the rotary encoder, we checked whether a timestamp has been assigned to each wheel movement, and whether any of the data sets contains only NaN values. The wheel data for all sessions included in this release passed our quality control checks, as indicated in the *Wheel* column of the [general QC spreadsheet](#).

Detailed wheel quality control outputs and the associated code can be found in the associated [github repository](#).

Video data

Video data were recorded from three separate cameras per session and analyzed using an automated pipeline. We provide an overview of the pipeline here, full details can be found in the IBL video hardware and software white paper (*The International Brain Laboratory, 2022c*). The data viewer at <https://viz.internationalbrainlab.org/app> can be used to visually inspect raw videos overlaid with pose estimation results from individual datasets.

The IBL ephys rig contains three cameras: a `LEFT` camera (1280x1024, 60 FPS) and a `RIGHT` camera (640x512, 150 FPS) that record the mouse's face, paws, and upper body from the side, and a `BODY` camera (640x512, 30 FPS) that records the mouse's body in the holder from above. We use DeepLabCut (DLC) (*Mathis et al., 2018*) to perform pose estimation on the three videos from each session (Fig 4). We also compute the motion energy of the whisker pads and the body, the times of lick events and the pupil diameter as additional behavioral features. Below we describe various automated quality control (QC) metrics computed on the raw videos and the DLC outputs.

247 The overall video and DLC QC outcomes for each camera can be found in the respective columns in the [general QC spreadsheet](#).
 248 Note that for some sessions the data for one or more cameras was not recorded or not released due to severe issues, and is
 249 indicated as MISSING. Otherwise, the values PASS, WARNING or FAIL indicate the worst of all QC checks for a given camera
 250 and data type.

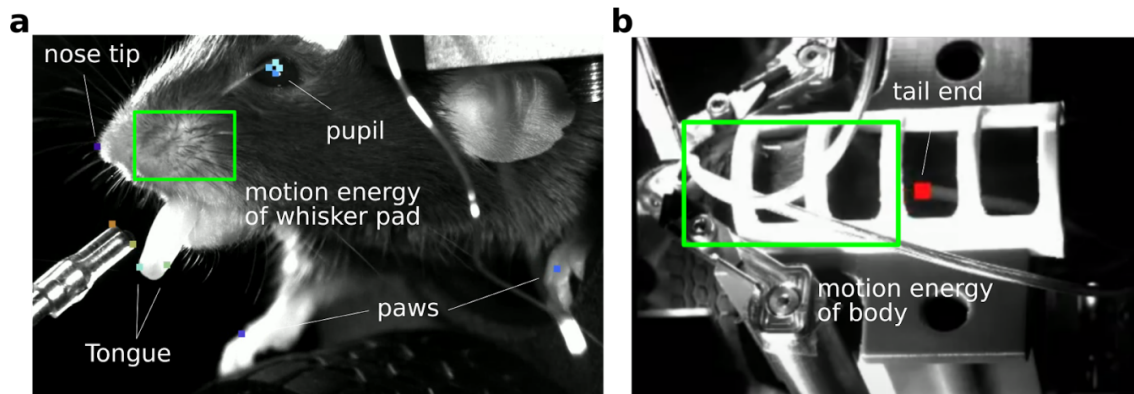


Figure 4. Video analysis pipeline outputs: **a.** Frame of left camera video with location of the points that are routinely tracked for LEFT and RIGHT videos. Green rectangular region indicates the whisker pad region used to compute motion energy. **b.** Frame of BODY camera video with the location of the tail end which is tracked in this view. Green rectangle indicates the region for which the motion energy of the animal's body is calculated.



Figure 5. Raw video QC examples from left camera: **a.** Example frame from each of four different sessions that pass all raw video QC checks. **b.** Examples that fail the position check. **c.** Examples that fail the brightness check.

251 Raw video QC

252 We report a standardized set of QC metrics for each video, such as whether a timestamp can be associated with each frame
253 of the video; whether the mouse is positioned correctly in the camera frame; and whether the camera brightness is adequate.
254 Figure 5 displays example frames of `LEFT` videos from several sessions that either pass all checks or fail a specific check.

255 All raw video QC metrics are defined [here](#) and further described in the aforementioned IBL video white paper (*The International*
256 *Brain Laboratory, 2022c*). We also report an overall outcome for each video based on these metrics. Below are the number of
257 videos from each of 459 sessions that have an overall QC outcome of `PASS`/`WARNING`/`FAIL`/`CRITICAL`/`MISSING`:

- 258 • `RIGHT`: 86/323/24/20/6
- 259 • `LEFT`: 94/293/49/18/5
- 260 • `BODY`: 33/217/7/196/6

261 Many of the `WARNING`/`FAIL` checks result for videos that are not positioned ideally or are too dark; these videos can
262 nevertheless be run through the pose estimation pipeline (described below) and often lead to reasonable behavioral features
263 (see for example top panels of Fig 6), which have their own set of QC metrics. The `CRITICAL` checks stem from issues in the
264 alignment between the video data and the behavioral data, or have otherwise been manually labeled as unsuitable for further
265 analysis. While these videos can also be run through pose estimation, aligning the resulting behavioral features to trial-based
266 events or neural activity should be performed with great care. Details on the outcomes for each QC check can be found in the
267 [general QC spreadsheet](#).

268 Pose estimation QC

269 We report a standardized set of QC metrics for the DLC traces, such as a check that the length of the DLC traces matches the
270 length of the video. All DLC QC metrics are defined [here](#) and their outcomes can be found in the [general QC spreadsheet](#).

271 **Paws.** Paw movements feature prominently in several IBL analyses, and as such we provide a specific QC metric for paw tracking.
272 For a given paw in one video, we compute the fraction of time points where paw likelihood falls below a given threshold in a
273 window of data aligned to stimulus onset for each trial (Fig 6). This fraction is then used to assign `PASS`/`FAIL`/`NOT_COMPUTED`
274 to the tracking of each paw in each video. This procedure is performed for both paws separately.

- 275 • `RIGHT` close paw: 336/8/115
- 276 • `RIGHT` far paw: 269/75/115
- 277 • `LEFT` close paw: 348/10/101
- 278 • `LEFT` far paw: 301/57/101

279 Overall tracking is better for the paw closest to the camera (“close paw” above), as the far paw is more often occluded by the body
280 or experimental equipment. Many of the `FAIL` cases reported here are due to underlying raw video issues (position, brightness,
281 etc.).

282 **Pupil.** We also provide an estimate of the pupil diameter from the `LEFT` and `RIGHT` cameras. Pupil diameter is estimated
283 using the top, bottom, right and left pupil markers after dropping low-likelihood markers. The resulting 1-dimensional trace
284 y is noisy and often contains missing data. We compute a smoothed version of this trace $\hat{y}$ using a Savitzky-Golay filter that
285 interpolates through missing data. Both y and $\hat{y}$ are made available. We estimate the signal-to-noise ratio (SNR) by computing
286 $SNR = \text{var}(\hat{y}) / \text{var}(\hat{y} - y)$ (code can be found [here](#)). An SNR threshold is used to assign `PASS`/`FAIL`/`NOT_COMPUTED` to each
287 pupil diameter estimate.

- 288 • `RIGHT`: 155/286/18
- 289 • `LEFT`: 157/297/5

290 The pupil diameter is not well tracked in our current pipeline, often due to the quality of the raw videos and/or the presence of
291 moving whiskers and infrared light reflections on the eye (Fig 7). Pupil diameter estimates are somewhat more reliable from
292 the `LEFT` camera, which has higher spatial resolution than the `RIGHT` camera. We are working to improve pupil diameter
293 estimation from the video data (*Biderman et al., 2024*).

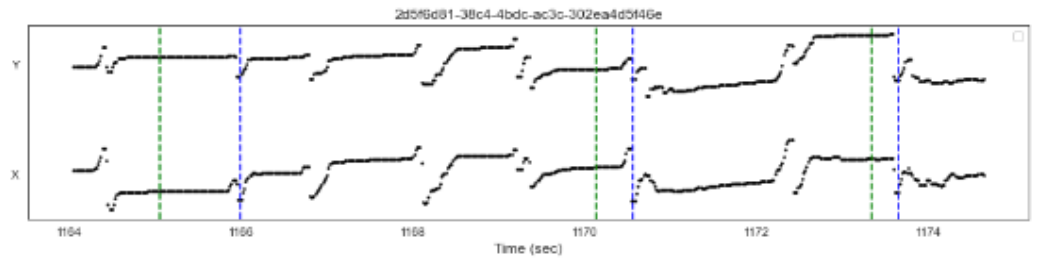
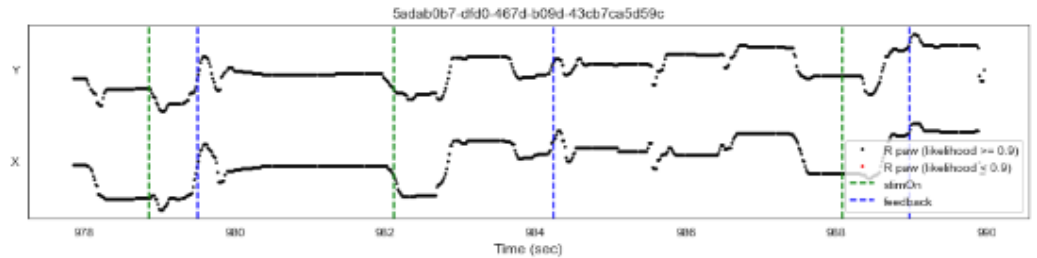
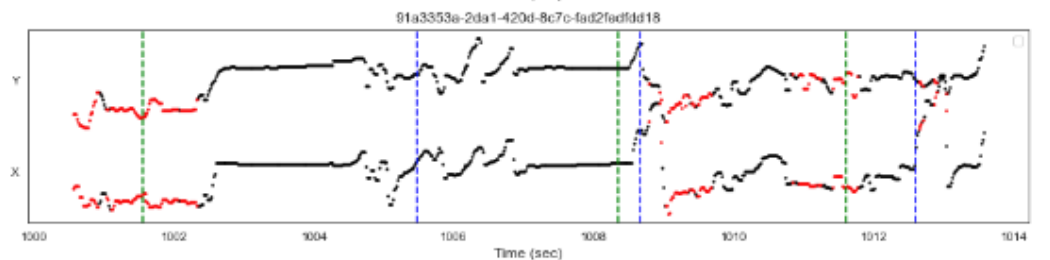
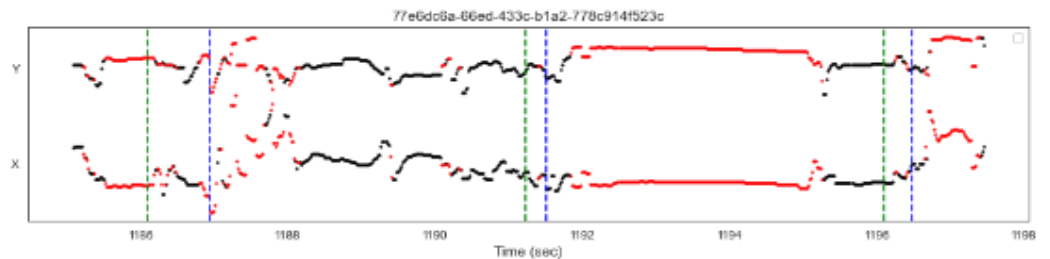
a**Pass paw QC check****b****Fail paw QC check**

Figure 6. Paw QC examples from left camera: a. Example frame and traces from each of two different sessions that pass paw QC check. Top line is paw y-coordinate, bottom line is paw x-coordinate; high likelihood points (likelihood ≥ 0.9) are plotted in black, low likelihood points (< 0.9) are plotted in red. Green dashed vertical lines indicate stimulus onset, blue dashed vertical lines indicate feedback onset. **b.** Examples that fail the paw QC check.

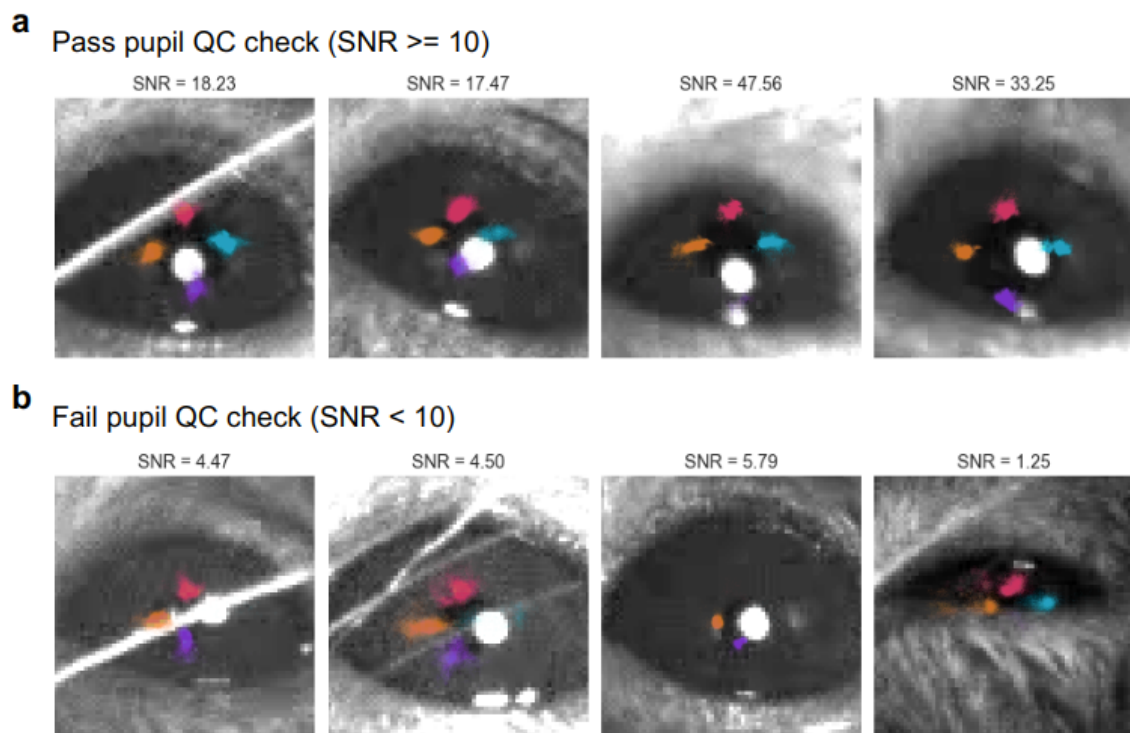


Figure 7. Pupil QC examples from left camera: a. Example frame from each of four different sessions that pass pupil QC check, overlaid with tracked points across the session. **b.** Examples that fail the pupil QC check. Frames have been brightened to increase pupil visibility.

Electrophysiology data

We visually inspected the raw data as well as the spike sorting output, and labelled them as `PASS`, `WARNING` or `FAIL` based on criteria defined below. Overall, 92 % of the raw ephys data was marked as `PASS`, 6 % as `WARNING` and the remaining 1 % judged as `FAIL` was not released. For the spike sorting output however, we decided to release the severe cases as the raw data appears of reasonable quality. 62 % of the spike sorting output was marked as `PASS`, 17 % as `WARNING`, and 20 % as `FAIL`.

Details can be found in the [online table](#) , and in the [general QC spreadsheet](#).

Raw ephys data QC

Specifically, we inspected 40ms snippets of raw data taken at three different time points in the recording : `T=600`, `1800` and `3000` seconds. The raw data was plotted using a gray-scale along the probe depth (example Fig 8); dark and white colors indicate a shift from baseline. Spikes detected were overlaid (good units in green, and multi-units in red). Brain regions were displayed along the probe depth next to the gray image, to determine whether noisy-presenting channels were indeed inside the brain and should be considered as such.

When inspecting the raw data, we highlighted these issues:

- Recordings with noisy channels (example Fig 9, where the top channels are noisy)
- Recordings with brief regular line noise artifacts (example Fig 10, see the vertical stripy bands appearing regularly).
- Recordings with brief regular artefacts appearing on part of the channels (example Fig 11, channels 140-160)
- Recordings with brief artifacts spanning the whole probe (example Fig 12 at `T=523s`)

Recordings presenting such defaults are all marked as `WARNING` in the associated [general QC spreadsheet](#) (column "Quality (Raw data)"). More severe cases were removed from the data release.

Spike sorting QC

When checking the spike sorting output, we looked both at the matching between the raw data and the spikes detected, and at the spikes raster overall (see Fig 13 for an example raster). We highlighted these issues, of which we show severe cases as examples:

- Recordings with a sharp discontinuity in the raster ; labelled as `Discontinuity`
 1. Recordings that appear stable, but have a sudden loss of spikes (example Fig 14)
 2. Recordings that appear stable, but have sudden appearance of spikes (example Fig 15)
Note: In some cases the raster presents major discontinuities, but the raw data appears stable throughout the recording (example Fig 20 and Fig 22)
 3. Recordings that appear stable, but have sudden horizontal artifacts (example Fig 16) ; also labelled as `Horizontal bands`
- Recordings with "smooth" loss / gain of spikes over time (example Fig 17) ; labelled as `Loss of spikes`
- Recordings with jittery-looking raster (example Fig 18) ; labelled as `Jitter`
Note: Often these recordings have stable raw data throughout the recording, but the spike detection does not seem to capture the underlying activity seen in the raw data (example Fig 18 and Fig 19)
- Recordings presenting an abnormal band of spikes, not obviously linked to the activity present in the raw data (example Fig 23-25) ; labelled as `Horizontal bands`
Note: This issue is likely due to the whitening process of the spike sorting algorithm.

Insertions presenting such issues are marked as `WARNING` or `FAIL` in the associated [general QC spreadsheet](#) (column "Quality (Spike sorting)"), depending on their severity. Insertions presenting such issues in a very mild manner, or not at all, were marked as `PASS` otherwise.

We also have noticed issues with the spike detection, that we did not label systematically:

- Cases of putative single spikes being detected several times (example Fig 26)

- Cases where the raw data presents activity, but the spikes detected are nearly null (example Fig 27)

In short, it appears that the current spike sorting pipeline has considerable room for improvement, and users of this data should keep these issues in mind and use the data viewer at <https://viz.internationalbrainlab.org/app> to check individual datasets as needed. We hope to correct these issues in future data releases.

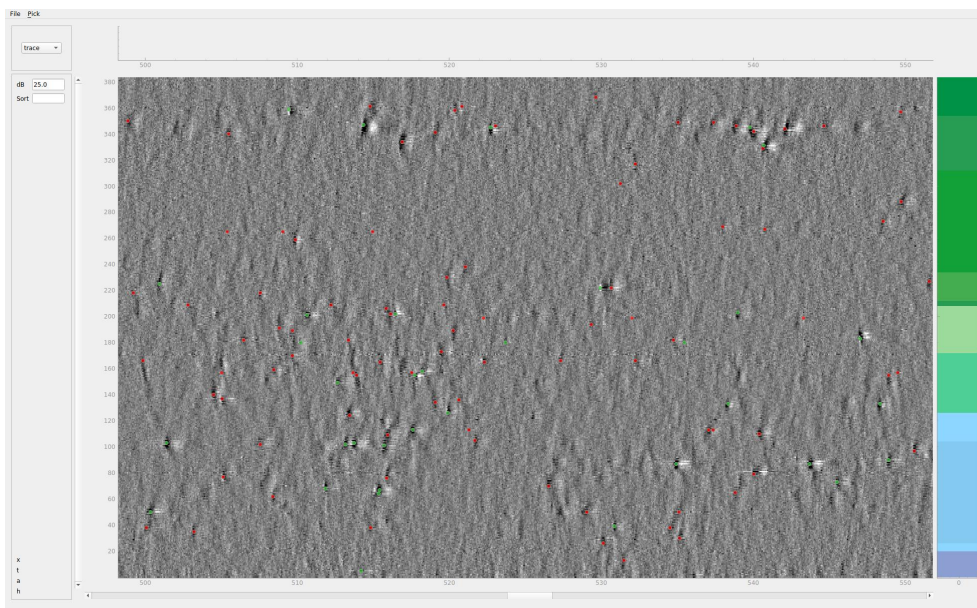


Figure 8. Example of good raw data. We present over time (x-axis) 40ms of raw data, displayed in a gray-scale for each channels on the probe (y-axis). The colors on the right indicate the brain regions crossed by the channels (using the Allen nomenclature). PID: 1bf5c05b-3654-4d5c-bac9-2b96edd12adf. Data displayed at T=1800s in recording.

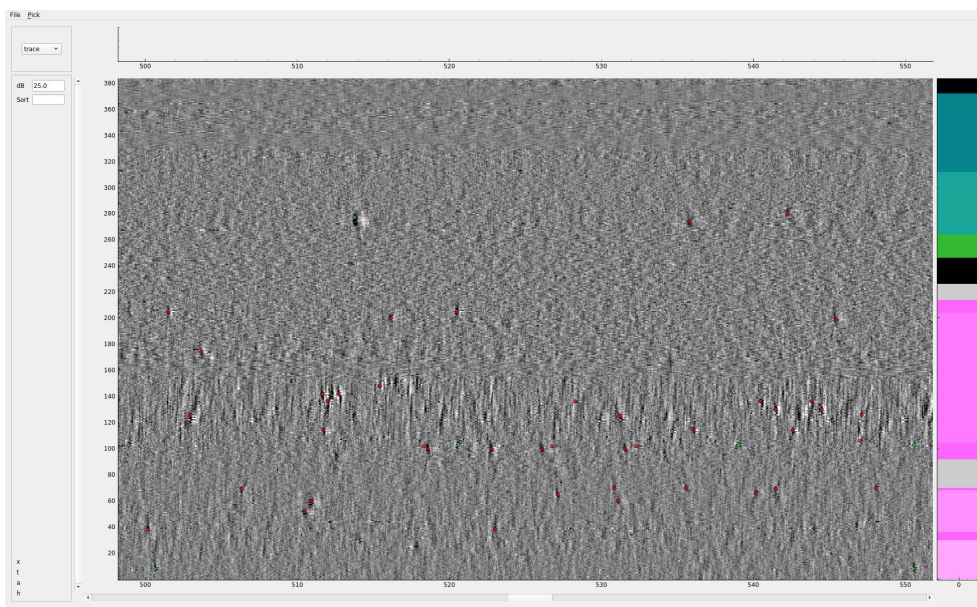


Figure 9. Example of noisy raw data, here where channels at the top (in cortical layers) appear as "blurred". PID: 578ca001-8cf5-4d30-b58f-1d574eaf808a. Data displayed at T=600s in recording.

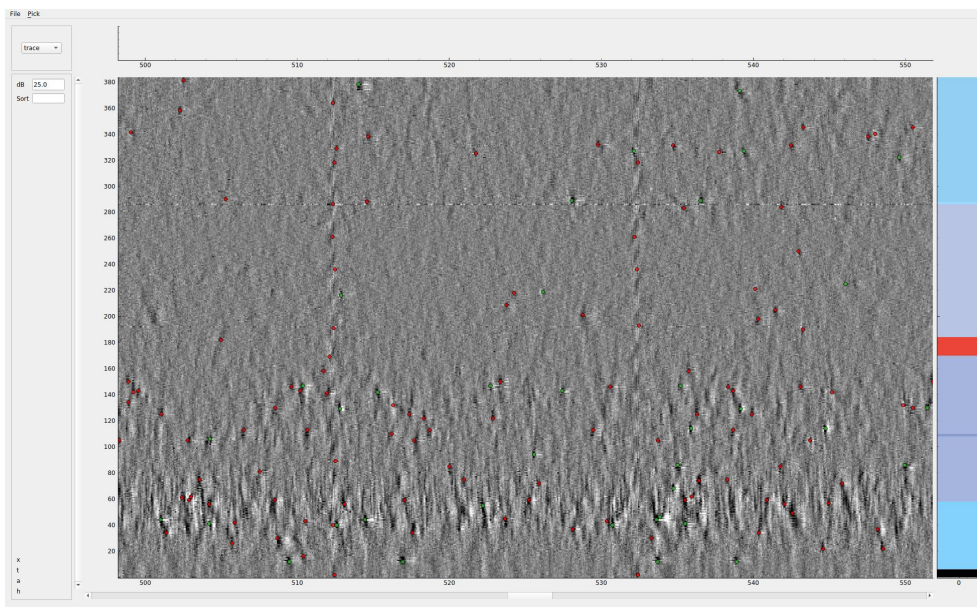


Figure 10. Example of noisy raw data, here displaying line noise. See the vertical stripy bands appearing regularly at a frequency nearing 50Hz. PID: a5f2ec22-0ff3-4249-bd2f-6247c3990e53. Data displayed at T=600s in recording.

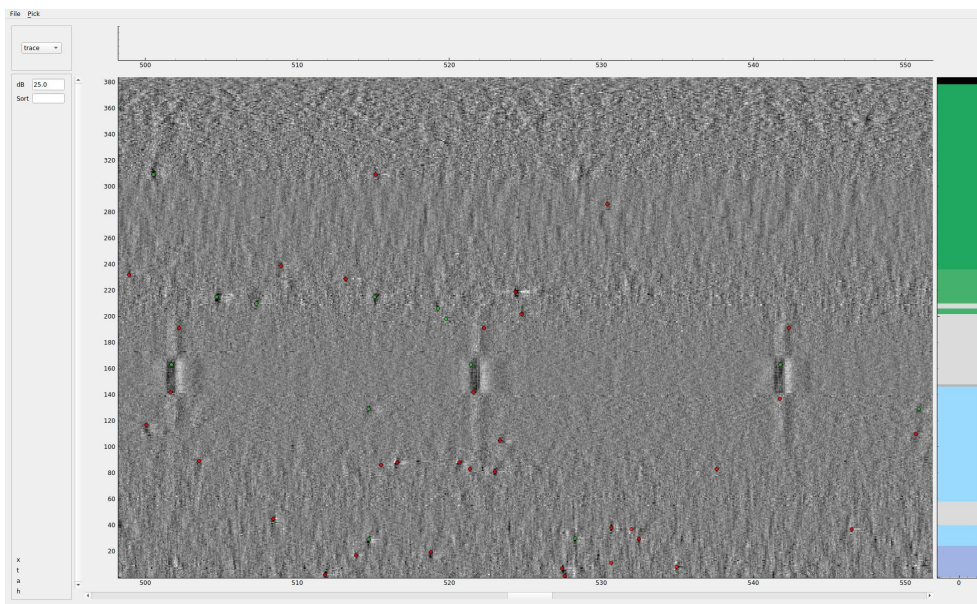


Figure 11. Example of noisy raw data, here displaying artifacts present only on some channels. See the "square" artifact visible on channels 140-160, appearing at regular (line noise) intervals. PID: af2a0072-e17e-4368-b80b-1359bf6d4647. Data displayed at T=600s in recording.

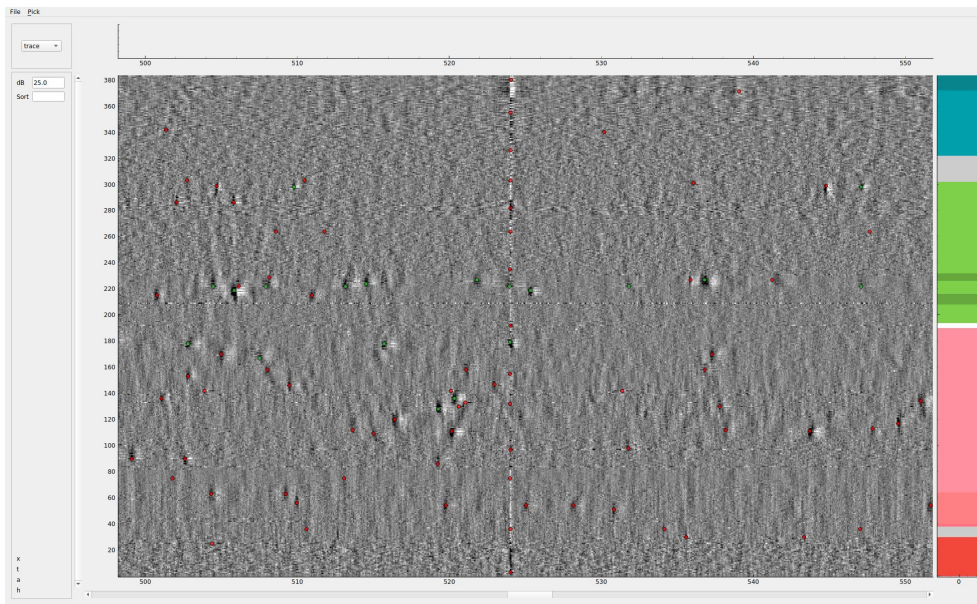


Figure 12. Example of noisy raw data, here presenting a brief artifact spanning the whole probe. See the vertical line at $x=523$. PID: c07d13ed-e387-4457-8e33-1d16aed3fd92. Data displayed at $T=600s$ in recording.

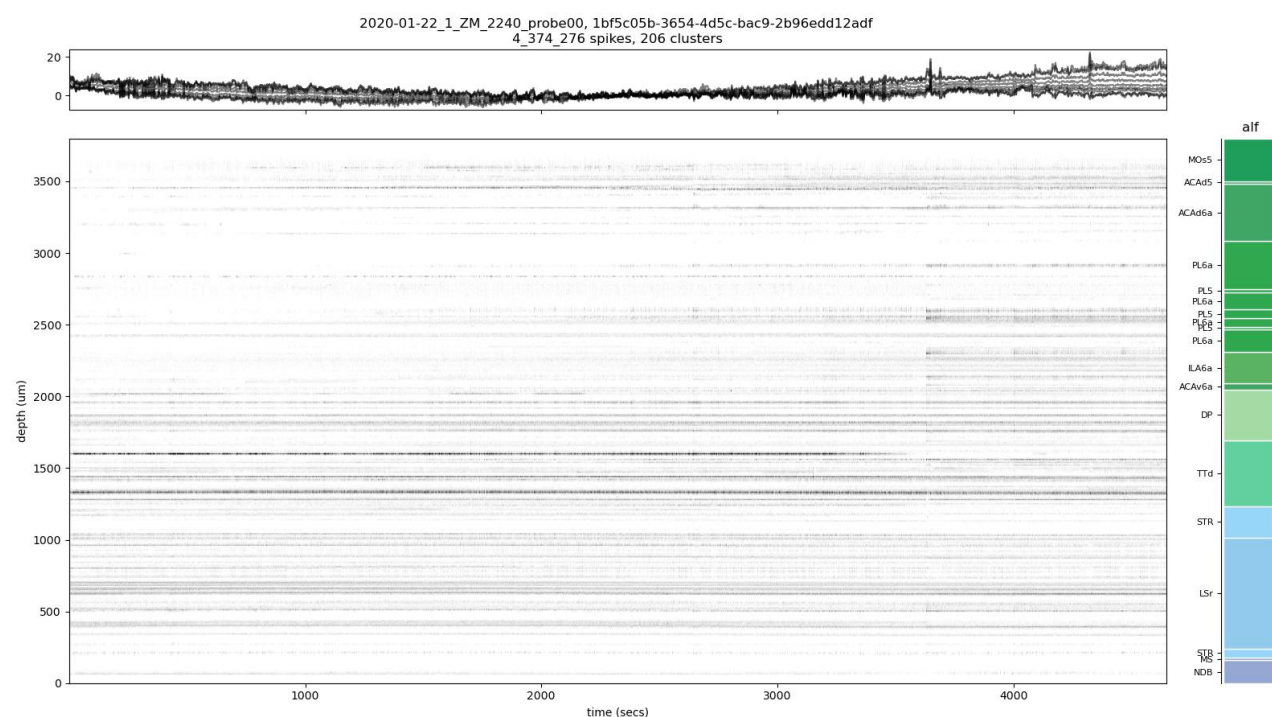


Figure 13. Example of good raster (good units). On top of the raster, we show the drift correction applied as part of the spike sorting processing (non-rigid drift correction applied here; each trace corresponds to the drift correction applied to a spatial subset of the probe). On the right, the brain regions crossed at labelled (as per the Allen nomenclature). PID: 1bf5c05b-3654-4d5c-bac9-2b96edd12adf

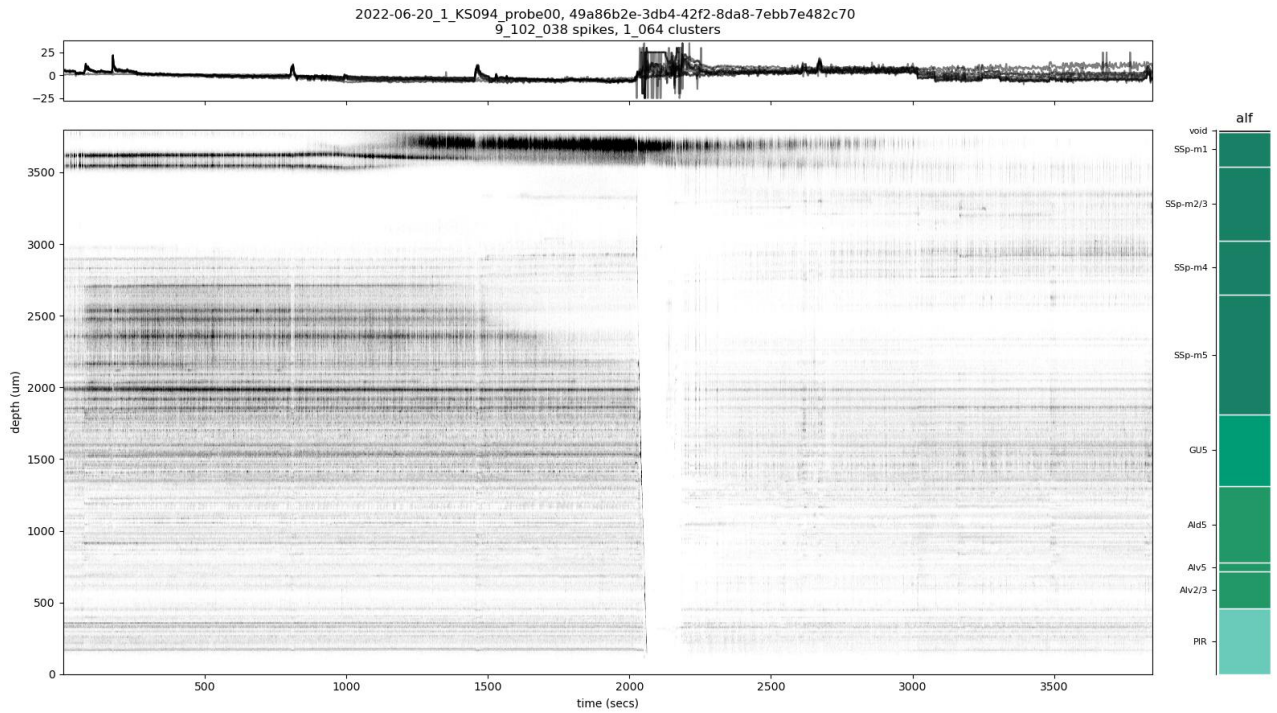


Figure 14. Example of a raster presenting a discontinuity, here a sudden loss of spikes (all units). See the gap in spikes after T=2000s. PID: 49a86b2e-3db4-42f2-8da8-7ebb7e482c70

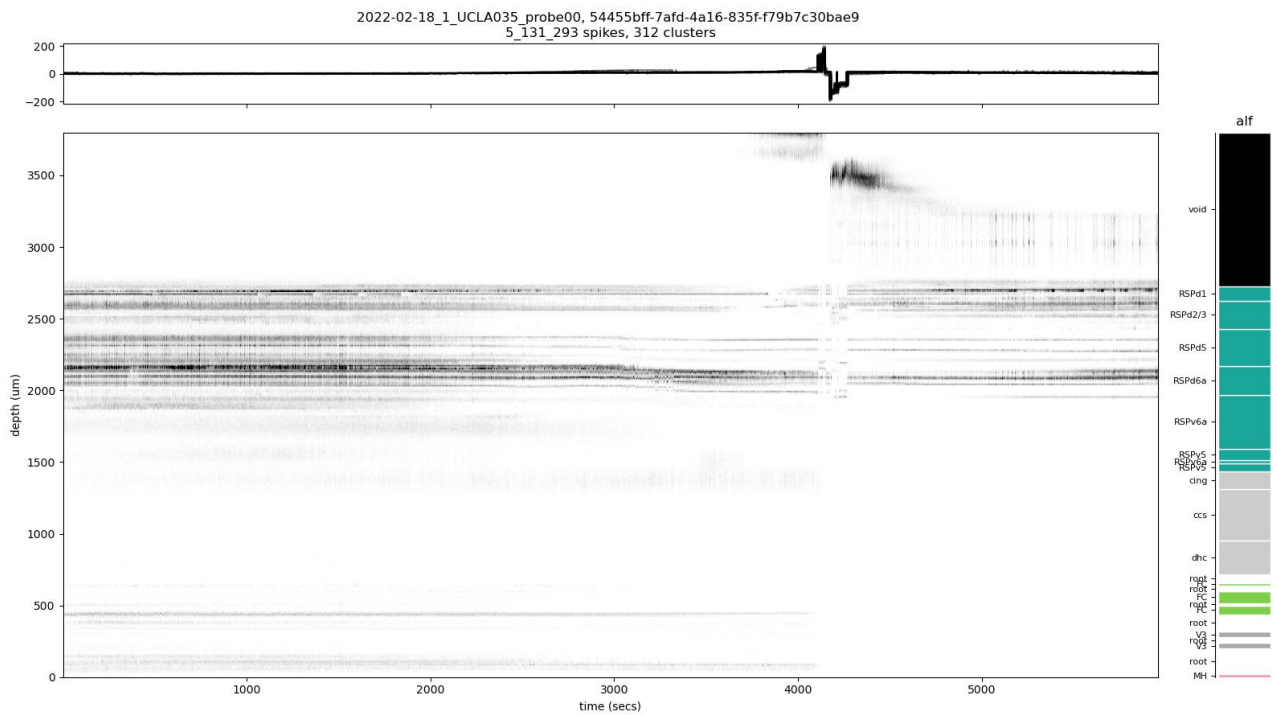


Figure 15. Example of a raster presenting a discontinuity, here a sudden gain of spikes (all units). See the loss and rebound in spikes after T=4000s, especially near depth=3500 μm. PID: 54455bff-7afd-4a16-835f-f79b7c30bae9

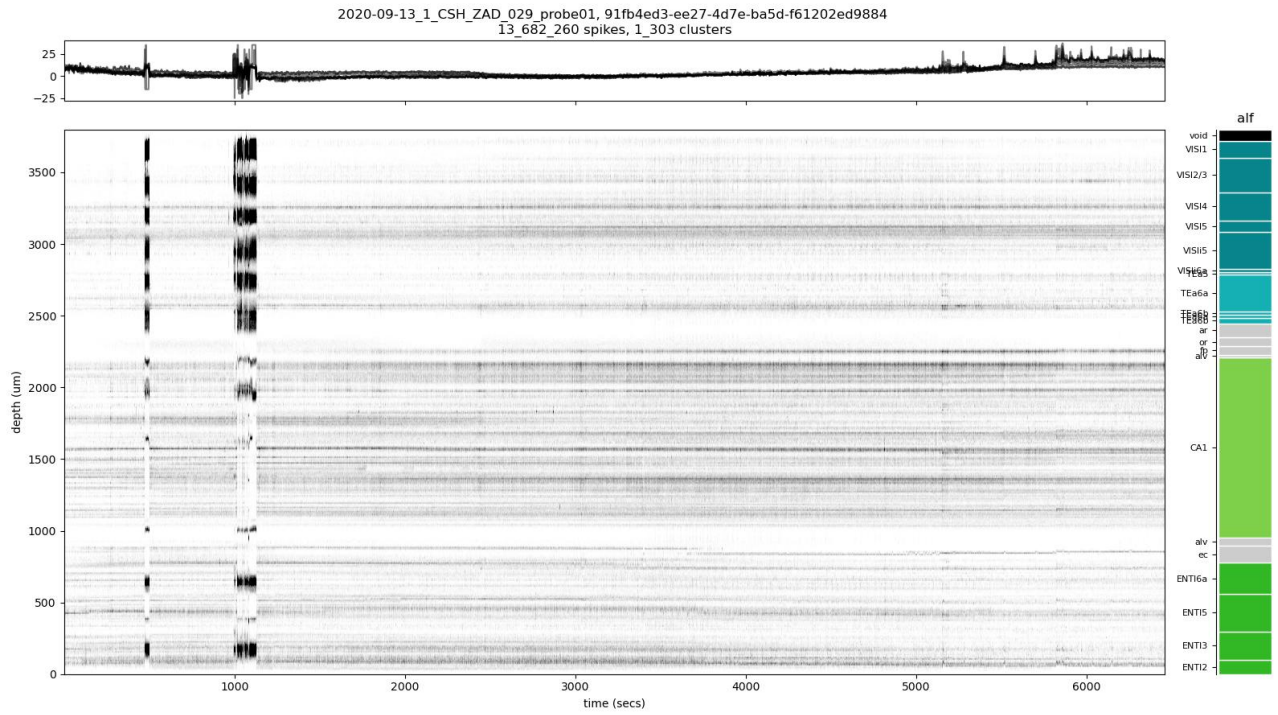


Figure 16. Example of a raster presenting a discontinuity, here sudden horizontal bands (all units). PID:

91fb4ed3-ee27-4d7e-ba5d-f61202ed9884

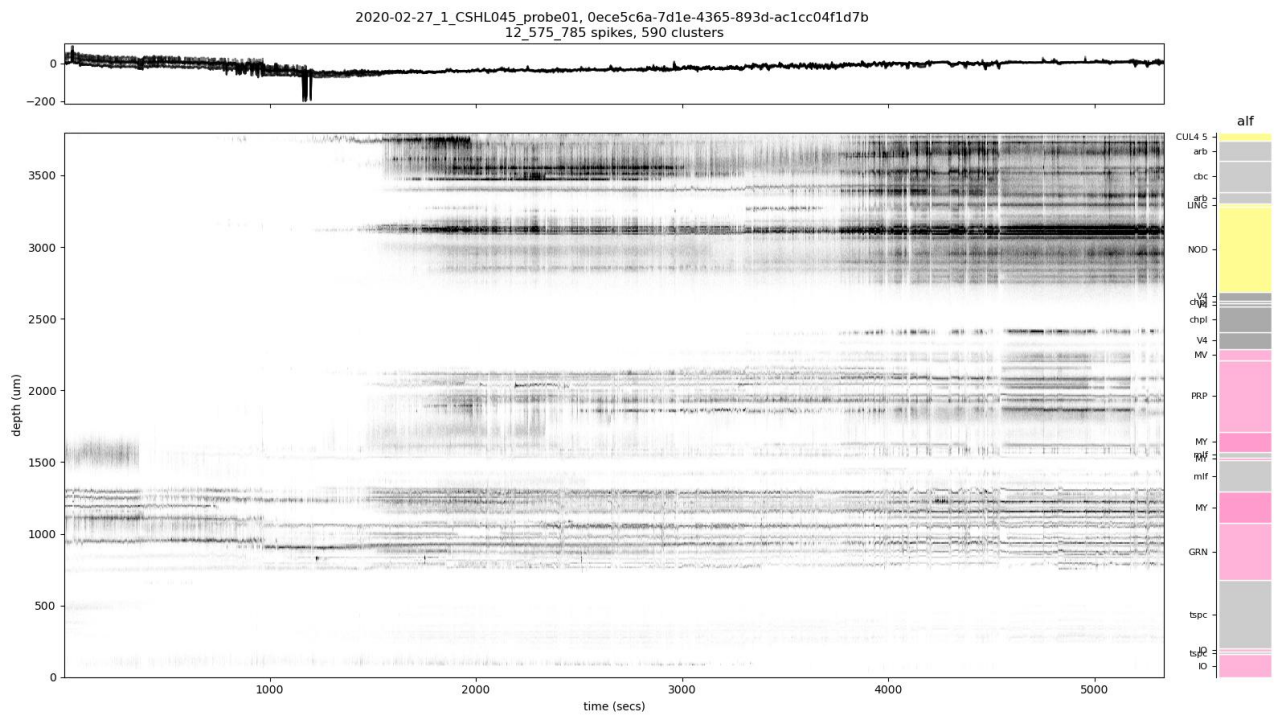


Figure 17. Example of a raster presenting a discontinuity, here a smooth gain in spiking activity over time, visible particularly near depth=2000-3500 μm (all units). PID:

0ece5c6a-7d1e-4365-893d-ac1cc04f1d7b

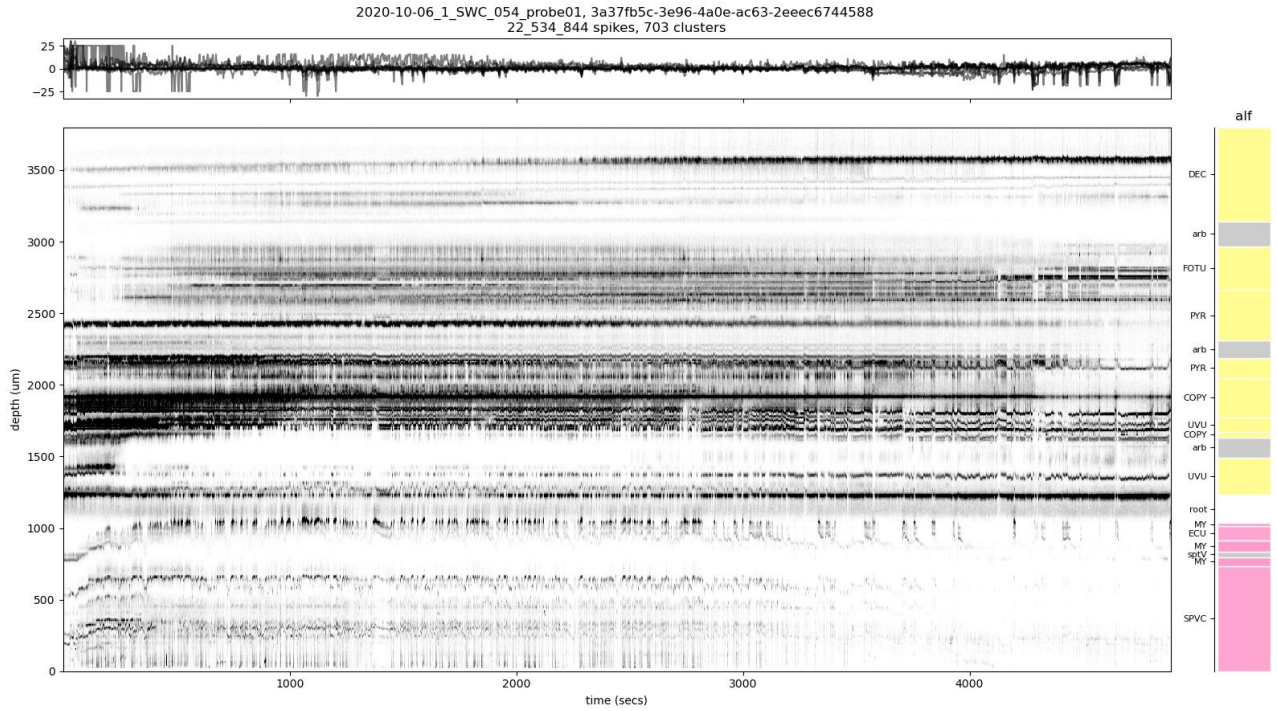


Figure 18. Example of a raster presenting a discontinuity, here a severe case of jitter (all units). See for example near depth=1000 μm . PID: 3a37fb5c-3e96-4a0e-ac63-2ecec6744588

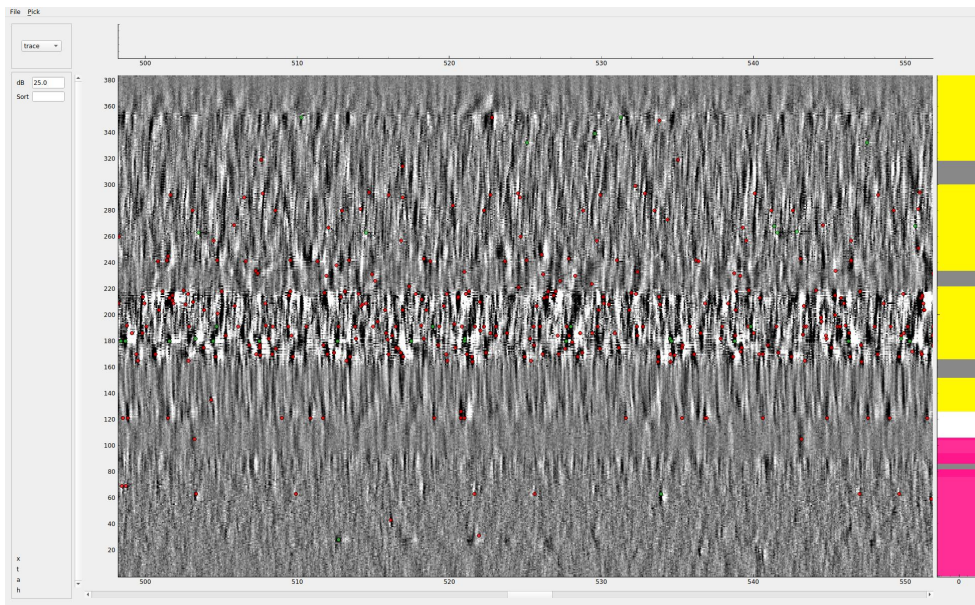


Figure 19. Example of raw data in the case of a jittery-looking raster. The raw data presents highly-active patterns, some of which is not caught by the spike detection. PID: 3a37fb5c-3e96-4a0e-ac63-2ecec6744588, same as in Fig fig:3a37fb5c_{aster}. Datadisplayedat $T = 600\text{sinrecording}$.

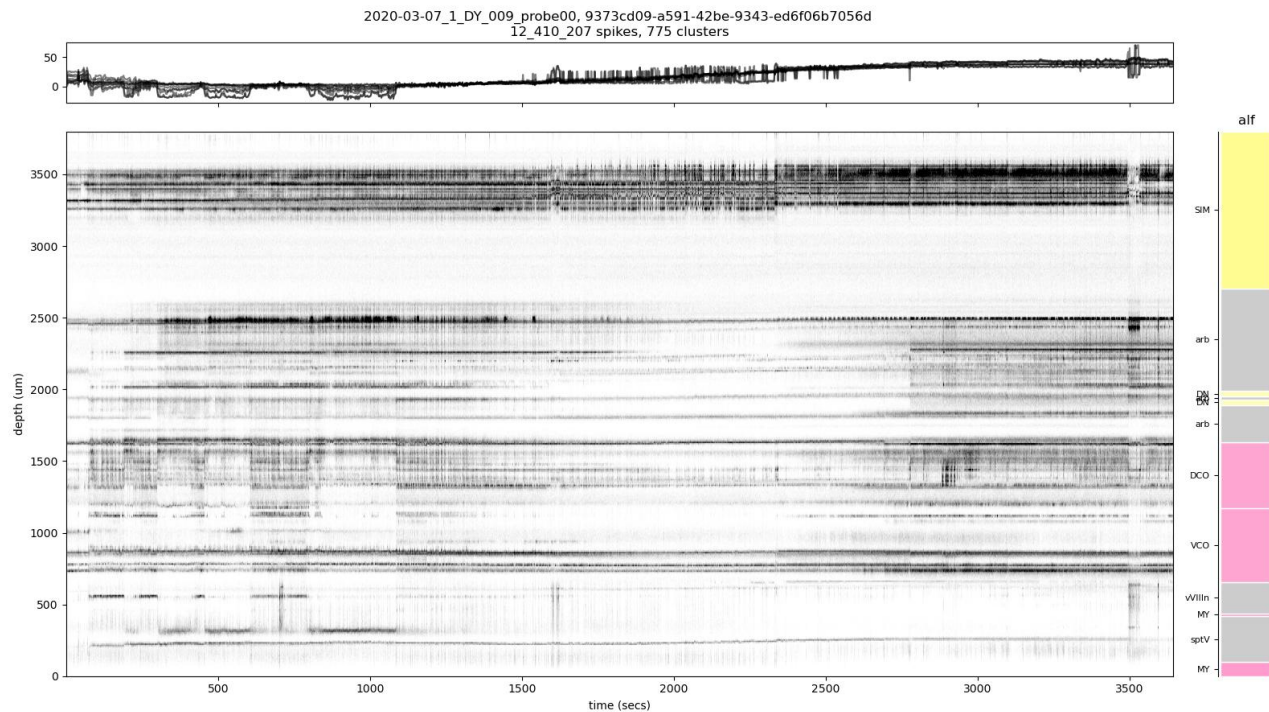


Figure 20. Example of a raster presenting discontinuities, but the raw data appears stable (all units). For example, at time T=600s, there is a clear diminution of activity compared to e.g. at time T=1250s or T=3000s. These sharp changes are not observed in the raw data. See the corresponding figures below, showing raw data at these two extreme time points (600s ; 3000s). PID:

9373cd09-a591-42be-9343-ed6f06b7056d

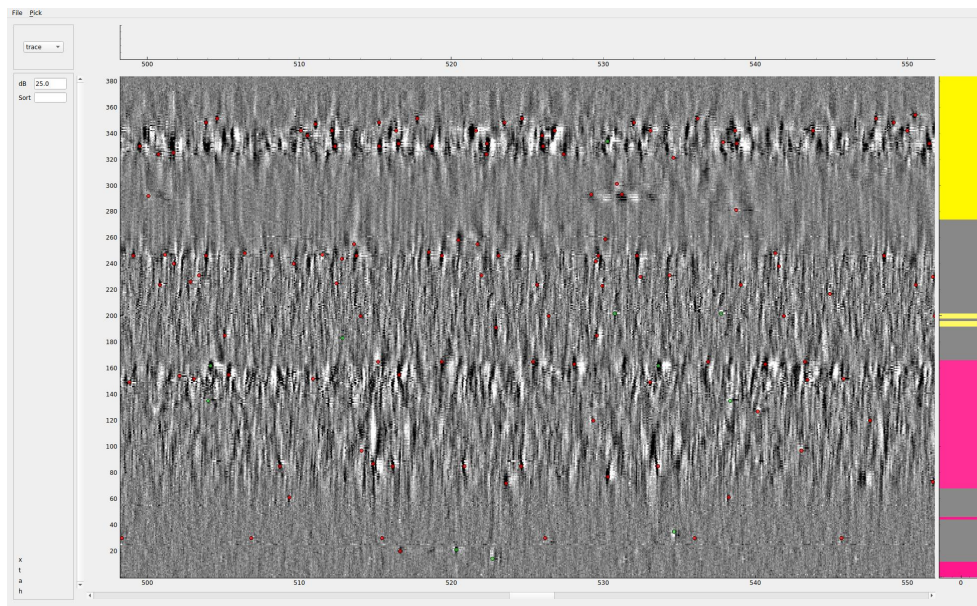


Figure 21. Example snippet of raw data at a time where we see a clear "dip" in the raster of Fig 20. PID:

9373cd09-a591-42be-9343-ed6f06b7056d. Data displayed at T=600s in recording.

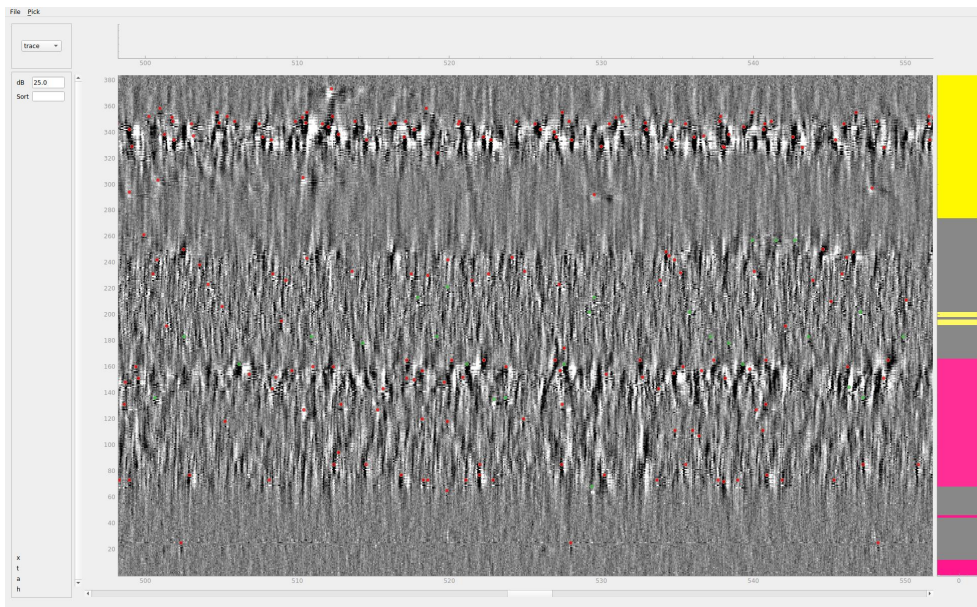


Figure 22. Example snippet of raw data at a later time in the raster of Fig 20. The activity is comparable to that seen in Fig 22. PID: 9373cd09-a591-42be-9343-ed6f06b7056d. Data displayed at T=3000s in recording.

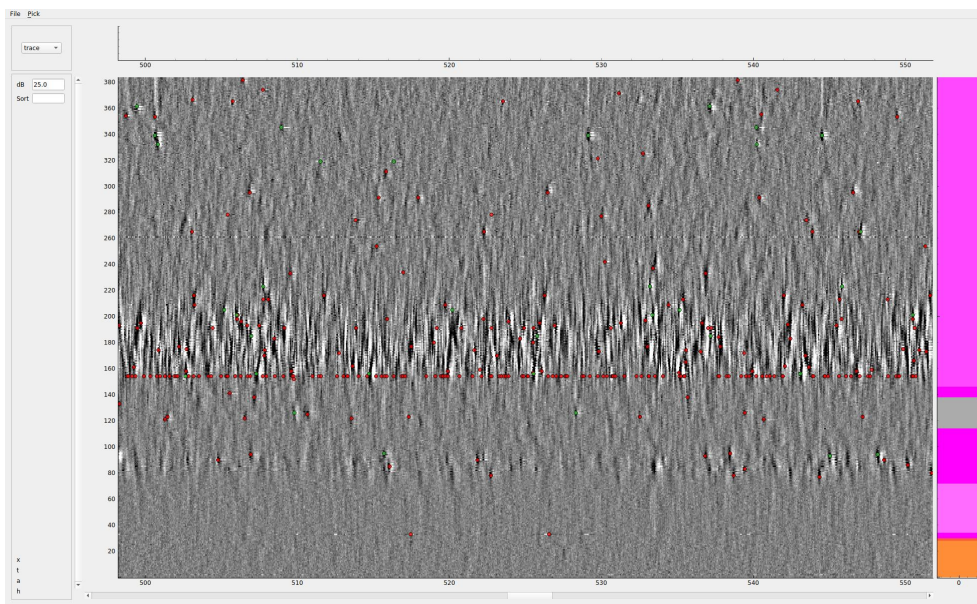


Figure 23. Example of raw data with abnormal spike detection. See the band of red (multi-units) spikes detected near channel 155. These spikes are not correlated with any activity seen on the raw data. PID: 3c283107-7012-48fc-a6c2-ed096b23974f. Data displayed at T=600s in recording.

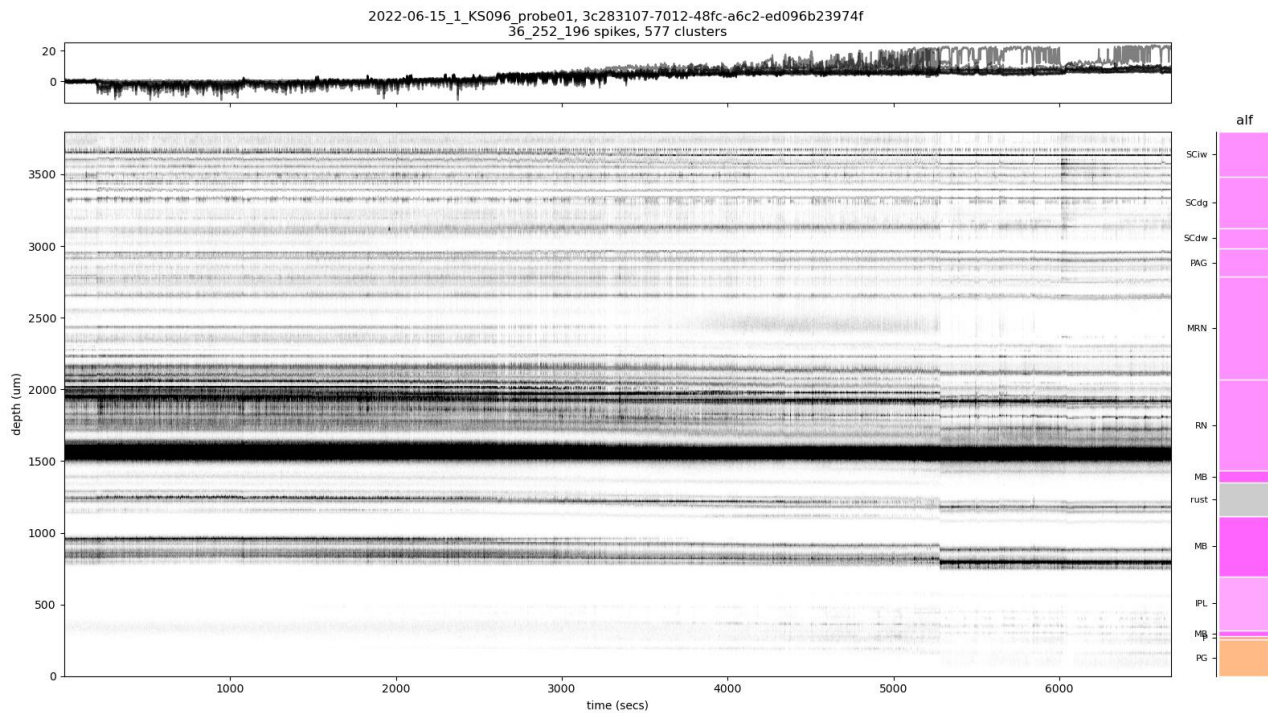


Figure 24. Example of raster with abnormal spike detection (all units). See the dark band near depth 1500μm, emerging from the poor spike detection visible from the raw data display in Fig 23. PID: 3c283107-7012-48fc-a6c2-ed096b23974f

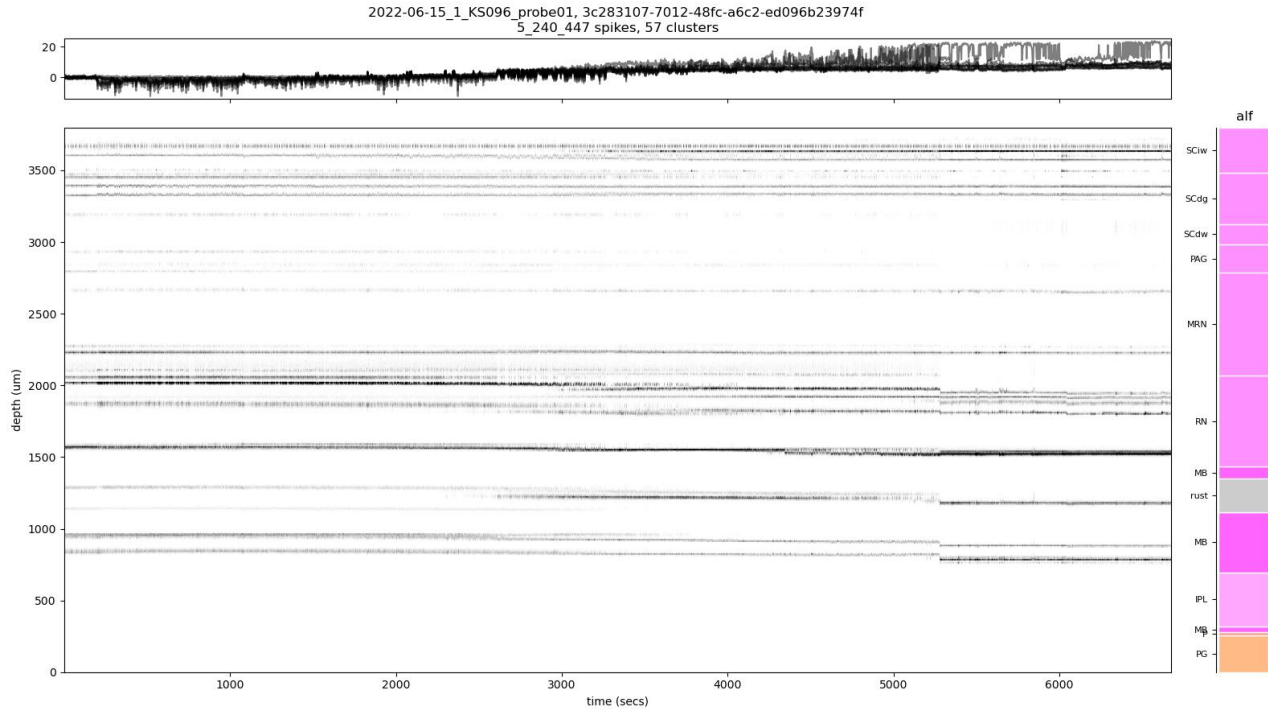


Figure 25. Example of raster with abnormal spike detection (good units). The dark band near depth 1500μm that appeared in Fig 24 is removed by the filtering for good units. PID: 3c283107-7012-48fc-a6c2-ed096b23974f

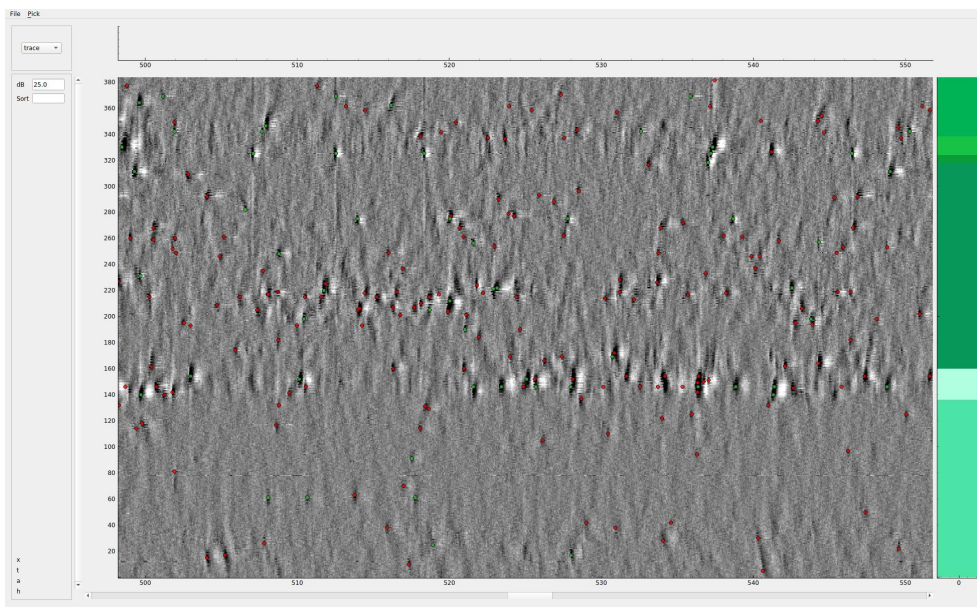


Figure 26. Example of raw data, with putatively single spikes appearing as being detected multiple times. See an example spike at $x=532, y=140$, with two green dots overlaid (these are spikes associated with good units). PID: df350b09-a420-42bb-9952-17862cbb9f1c Data displayed at T=600s in recording.

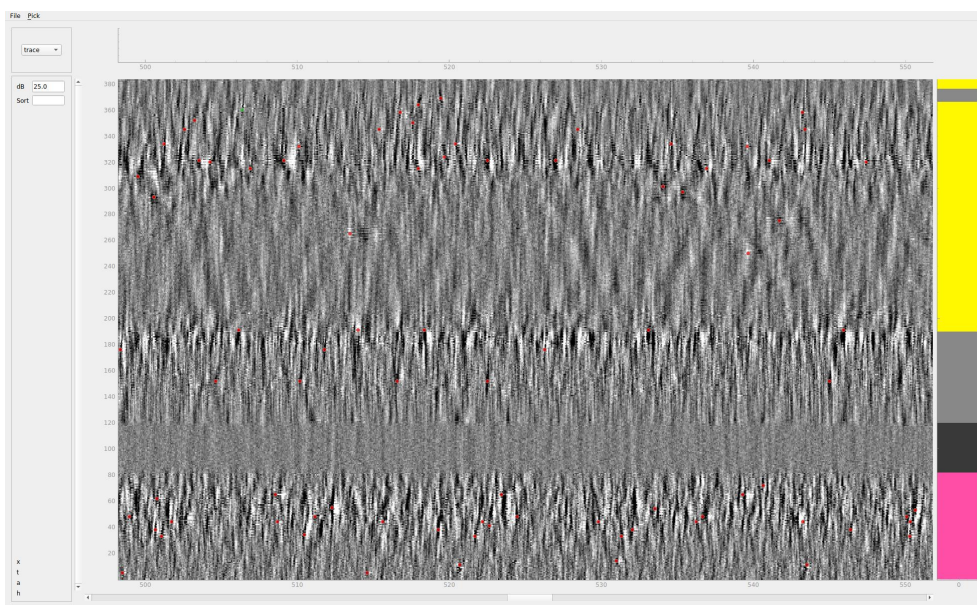


Figure 27. Example of raw data, with low spike detection compared to the activity present in the raw data. PID: 577e4741-4b15-4e91-b81b-61304a09bfb5. Data displayed at T=1800s in recording.

Acknowledgments

This work was supported by grants from the Wellcome Trust (209558 and 216324), National Institutes of Health (1U19NS123716, including a Diversity Supplement and an Administrative Supplement to Support Enhancement of Software Tools for Open Science) and the Simons Foundation.

Diversity Statement

We support inclusive, diverse and equitable conduct of research. One or more of the IBL members self-identifies as a member of an underrepresented ethnic minority in science. One or more of the IBL members self-identifies as a member of the LGBTQIA+ community.

References

- Biderman D**, Whiteway MR, Hurwitz C, Greenspan N, Lee RS, Vishnubhotla A, Warren R, Pedraja F, Noone D, Schartner MM, et al. Lightning Pose: improved animal pose estimation via semi-supervised learning, Bayesian ensembling and cloud-native open-source tools. *Nature Methods*. 2024; p. 1–13.
- Campbell R**, BakingTray; 2020. <https://github.com/SainsburyWellcomeCentre/BakingTray>, doi: <https://doi.org/10.5281/zenodo.3631609>.
- Campbell R**, StitchIt; 2021. <https://github.com/SainsburyWellcomeCentre/StitchIt>, doi: <https://zenodo.org/badge/latestdoi/57851444>.
- Campbell R**, Blot A, Rousseau C, Winter O, Lasagna; 2020. <https://github.com/SainsburyWellcomeCentre/lasagna>, doi: [10.5281/zenodo.3941894](https://doi.org/10.5281/zenodo.3941894).
- Economo MN**, Clack NG, Lavis LD, Gerfen CR, Svoboda K, Myers EW, Chandrashekar J. A platform for brain-wide imaging and reconstruction of individual neurons. *eLife*. 2016; 5(e10566). doi: [10.7554/eLife.10566](https://doi.org/10.7554/eLife.10566).
- Faulkner M**, Ephys Atlas GUI; 2020. <https://github.com/int-brain-lab/iblapts/tree/master/atlas electrophysiology>.
- Jun JJ**, Steinmetz NA, Siegle JH, Denman DJ, Bauza M, Barbarits B, Lee AK, Anastassiou CA, Andrei A, Aydın Ç, et al. Fully integrated silicon probes for high-density recording of neural activity. *Nature*. 2017; 551(7679):232–236.
- Klein S**, Staring M, Murphy K, Viergever MA, Pluim JPW. elastix: a toolbox for intensity based medical image registration. *IEEE Transactions on Medical Imaging*. 2010; 29(1):196–205. doi: [10.1109/TMI.2009.2035616](https://doi.org/10.1109/TMI.2009.2035616).
- Liu LD**, Chen S, Hou H, West SJ, Faulkner M, Economo MN, Li N, Svoboda K, the International Brain Laboratory. Accurate localization of linear probe electrode arrays across multiple brains. *eNeuro*. 2021; 8(6). doi: [10.1523/ENEURO.0241-21.2021](https://doi.org/10.1523/ENEURO.0241-21.2021).
- Lopes G**, Bonacchi N, Frazão J, Neto JP, Atallah BV, Soares S, Moreira L, Matias S, Itskov PM, Correia PA, et al. Bonsai: an event-based framework for processing and controlling data streams. *Frontiers in neuroinformatics*. 2015; 9:7.
- Lopes G**, Farrell K, Horrocks EA, Lee CY, Morimoto MM, Muzzu T, Papanikolaou A, Rodrigues FR, Wheatcroft T, Zucca S, et al. Creating and controlling visual environments using BonVision. *Elife*. 2021; 10:e65541.
- Mathis A**, Mamidanna P, Cury KM, Abe T, Murthy VN, Mathis MW, Bethge M. DeepLabCut: markerless pose estimation of user-defined body parts with deep learning. *Nature neuroscience*. 2018; 21(9):1281–1289.
- Ragan T**, Kadiri LR, Venkataraju KU, Bahlmann K, Sutin J, Taranda J, Arganda-Carreras I, Kim Y, Seung HS, Osten P. Serial two-photon tomography for automated ex vivo mouse brain imaging. *Nat Methods*. 2012; 9(3):255–8. doi: [10.1038/nmeth.1854](https://doi.org/10.1038/nmeth.1854).
- Rossant C**, Winter O, Hunter M, Huntenburg J, Faulkner M, Wells M, Steinmetz N, Harris K, Bonacchi N, Alyx; 2021. <https://github.com/cortex-lab/alyx>.
- Steinmetz NA**, Aydın C, Lebedeva A, Okun M, Pachitariu M, Bauza M, Beau M, Bhagat J, Böhm C, Broux M, Chen S, Colonell J, Gardner RJ, Karsh B, Kloosterman F, Kostadinov D, Mora-Lopez C, O’Callaghan J, Park J, Putzeys J, et al. Neuropixels 2.0: A miniaturized high-density probe for stable, long-term brain recordings. *Science*. 2021; 372(6539):eabf4588.

378 **The International Brain Laboratory**, IBL sorter; 2021. <https://github.com/int-brain-lab/ibl-sorter>.

379 **The International Brain Laboratory**, iblvideo; 2021. <https://github.com/int-brain-lab/iblvideo>.

380 **The International Brain Laboratory**. Standardized and reproducible measurement of decision-making in mice. eLife. 2021; 10:e63711. doi:
381 [10.7554/eLife.63711](https://doi.org/10.7554/eLife.63711).

382 **The International Brain Laboratory**. Reproducibility of in-vivo electrophysiological measurements in mice. bioRxiv. 2022; .

383 **The International Brain Laboratory**. Spike sorting pipeline for the International Brain Laboratory. figshare. 2022; doi:
384 [10.6084/m9.figshare.19705522](https://doi.org/10.6084/m9.figshare.19705522).

385 **The International Brain Laboratory**. Video hardware and software for the International Brain Laboratory. figshare. 2022; doi:
386 [10.6084/m9.figshare.19694452](https://doi.org/10.6084/m9.figshare.19694452).

387 **Wang Q**, Ding SL, Li Y, Royall J, Feng D, Lesnar P, Graddis N, Naeemi M, Facer B, Ho A, Dolbeare T, Blanchard B, Dee N, Wakeman W, Hirokawa KE,
388 Szafer A, Sunkin SM, Oh SW, Bernard A, Phillips JW, et al. The Allen Mouse Brain Common Coordinate Framework: A 3D Reference Atlas. Cell.
389 2020; 181(4):936–953.e20. doi: [10.1016/j.cell.2020.04.007](https://doi.org/10.1016/j.cell.2020.04.007).

390 **West SJ**, BrainRegister; 2021. <https://github.com/stevenjwest/brainregister>.